Rendu 1

Rakuten France — E-commerce Product Classification

Exploratory Data Analysis, Data Visualisation & Preprocessing

DataScientest · Data Scientist Sprint 2 · February 2026

Dataset	Rakuten France Multimodal Classification (ENS ChallengeData #35)
Training set	84,916 products · 27 categories
Test set	13,812 products (unlabelled)
Modalities	Text (designation + description) + Product image
Target metric	Weighted F1-score (macro)
Deliverable	Cleaned dataset ready for Rendu 2 modelling

1. Project Introduction

1.1 Context

This report addresses the **Rakuten France Multimodal Product Data Classification** challenge (ENS ChallengeData #35). The task is to assign one of 27 product category codes (*prdtypecode*) to e-commerce listings using three sources:

- **designation** — short product title, always present
- **description** — HTML-rich text, missing in ~35% of rows
- **Product image** — JPEG photograph, always present

Economic context: Automated classification enables better search, recommendation, and inventory management at scale.

Scientific context: Multimodal classification requiring fusion of NLP and computer vision — an active machine learning research area.

1.2 Objectives

- Understand dataset: volume, quality, class distribution, missing data patterns
- Produce ≥5 annotated visualisations with statistical validation
- Preprocess text and images ready for Rendu 2 modelling
- Save cleaned train/test sets as pickle files for downstream use

2. Data Understanding

2.1 Dataset Structure

File	Rows	Columns	Role
x_train.csv	84,916	designation, description, productid, imageid	Training features
y_train.csv	84,916	prdtypecode	Training labels
test.csv	13,812	designation, description, productid, imageid	Test features (unlabelled)
images/image_train/	84,916	JPEG files	Product photos (train)
images/image_test/	13,812	JPEG files	Product photos (test)

2.2 Key Observations

- 84,916 labelled training samples across **27 product categories**
- **designation** always present; **description** missing in **35.1%** of rows
- Text predominantly French with some German and English
- **HTML tags** present in description — stripped during preprocessing
- All 84,916 training images present (0 missing)

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Training shape : (84,916 × 4) features + (84,916 × 1) label
Missing values : description → 29,800 (35.1%) | all other columns → 0
Target classes : 27 (codes: 10, 40, 50, …, 2905)
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3. Visualisations & Statistical Analysis

3.1 Class Distribution (Fig 1)

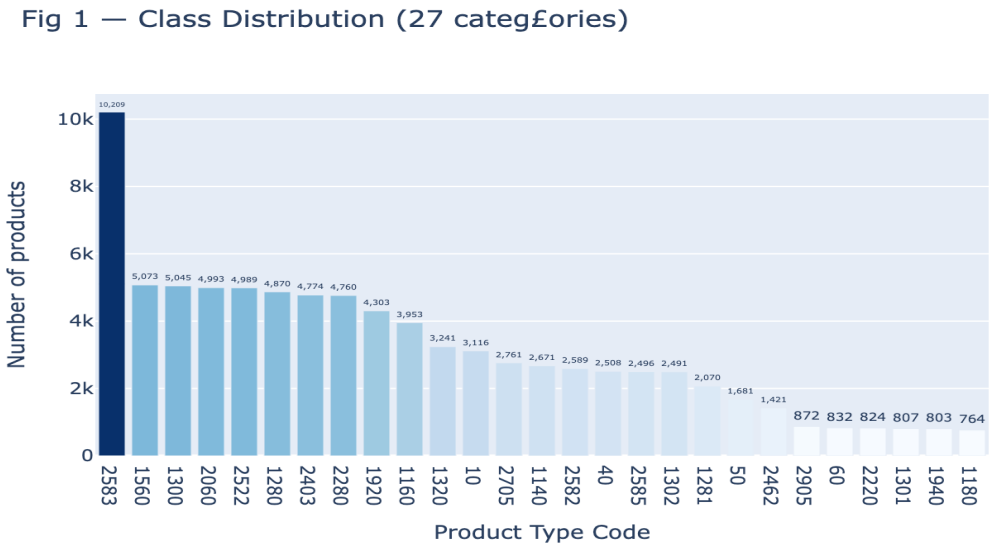


Fig 1 — Product count per category, sorted descending.

Business insight: The catalogue is heavily imbalanced. Category 2583 accounts for ~12% of all products (10,209 samples), while the smallest category 1180 has only 764 — a 13:1 ratio.

Statistical validation: Chi-squared goodness-of-fit vs uniform: χ^2 = 36,570.3, p ■ 0.05. Imbalance is statistically significant.

Recommendation:

- Apply class-weighted training (class_weight='balanced') to compensate for imbalance
- Use stratified train/validation splits to ensure all 27 classes appear in evaluation
- Monitor per-class F1 separately — weighted-F1 can mask poor performance on rare classes

3.2 Missing Description Rate per Category (Fig 2)

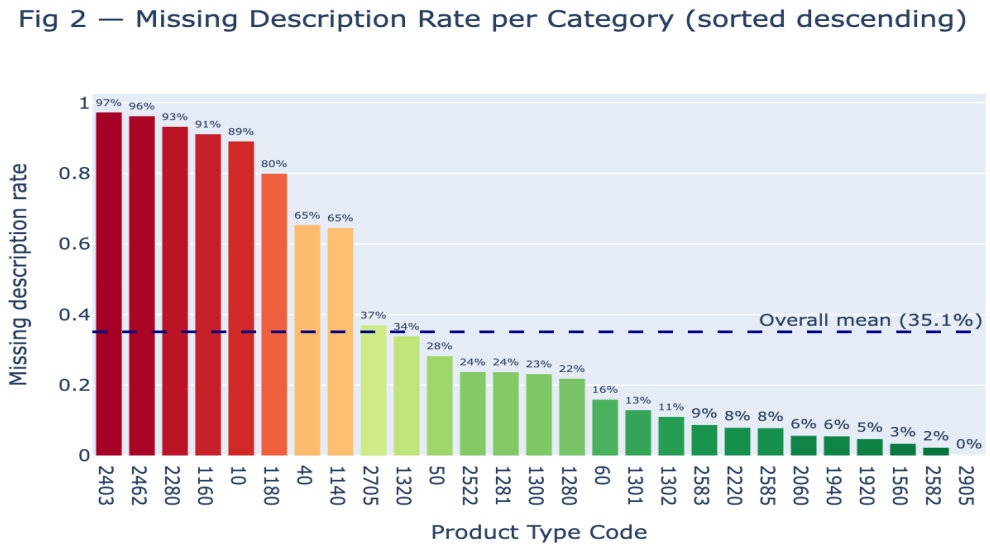


Fig 2 — Missing description rate per category, sorted descending.

Business insight: Missing rates vary from near 0% (books) to over 80% (video games, collectibles). High-missing categories are effectively image-only.

Statistical validation: Chi-squared test of independence (category × description presence): χ^2 = 43,757.8, df=26, p ■ 0.05.

Recommendation:

- Prioritise image features for categories with missing rate > 80%
- Fill missing descriptions with empty string — do not substitute designation
- Add has_description binary feature — missingness is a strong categorical signal

3.3 Missing Rate vs. Product Count (Fig 2b)

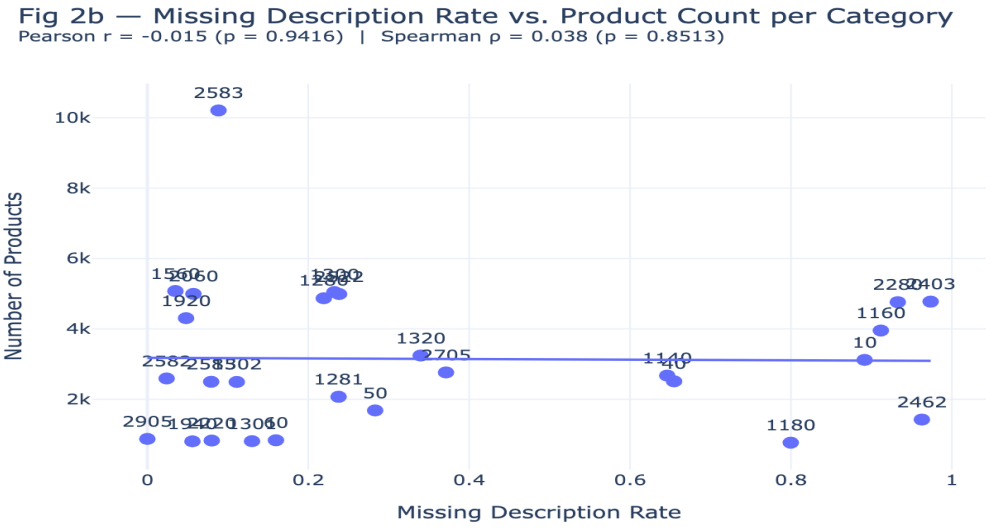


Fig 2b — Missing description rate vs. number of products per category. OLS trendline.

Business insight: Hypothesis: more missing descriptions → fewer products. Result: Pearson r = −0.015 (p = 0.94), Spearman ρ = 0.038 (p = 0.85) — not supported. Category size is driven by popularity, not description completeness.

Recommendation:

- Do not use missing description rate as a proxy for category size or data quality
- Treat each category's missingness pattern independently in the multimodal pipeline

3.4 Text Length Distribution per Category (Fig 3)

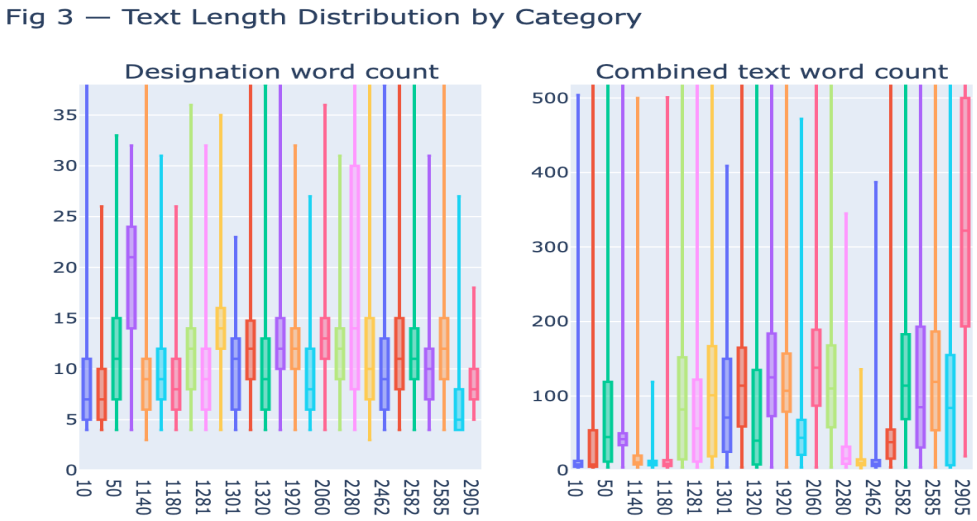


Fig 3 — Word count distribution: designation (left) and description (right).

Business insight: Text length varies significantly across categories. Books have long descriptions; other categories rely on short titles. Text length is itself an informative feature.

Statistical validation: Kruskal-Wallis test on combined word count: H = 34,696.6, p ■ 0.05.

Recommendation:

- Include word_count and char_count as explicit numeric features alongside embeddings
- Avoid aggressive truncation for description-rich categories
- Short-text categories may benefit more from image than text features

3.5 Image Sample Grid (Fig 4)

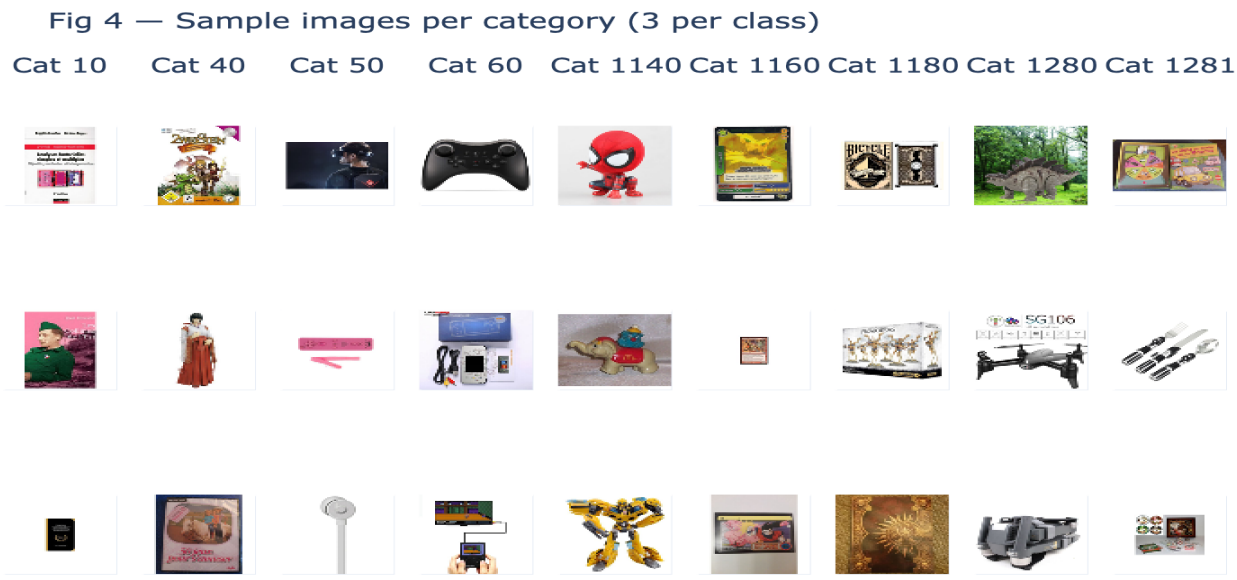


Fig 4 — Sample product images (3 per category) for 9 selected categories.

Business insight: Categories are often visually distinguishable, but within-category variation is high. Image features alone will not suffice — text signals are essential for disambiguation.

Recommendation:

- Apply data augmentation (random crop, flip, colour jitter) during CNN training
- Use late fusion: train text and image classifiers independently, then combine outputs
- For visually ambiguous categories, text features act as a tiebreaker

3.6 Top TF-IDF Terms per Category (Fig 5)

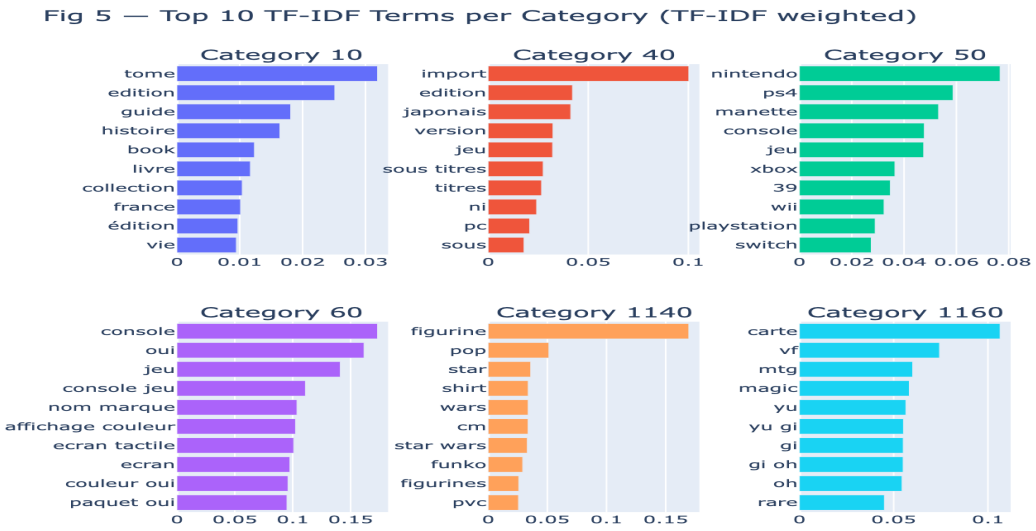


Fig 5 — Top 10 TF-IDF weighted terms for 6 sample categories.

Business insight: TF-IDF reveals strongly discriminative vocabulary per category. Cat 50 (video games): nintendo, ps4, manette, console. Cat 10 (books): tome, edition, guide, histoire.

Recommendation:

- Use top TF-IDF terms as a sanity check for label quality
- CamemBERT / FlauBERT will outperform multilingual models for French text
- Consider category-specific vocabulary masks to boost discriminative terms

3.7 Image Size Distribution (Fig 6)

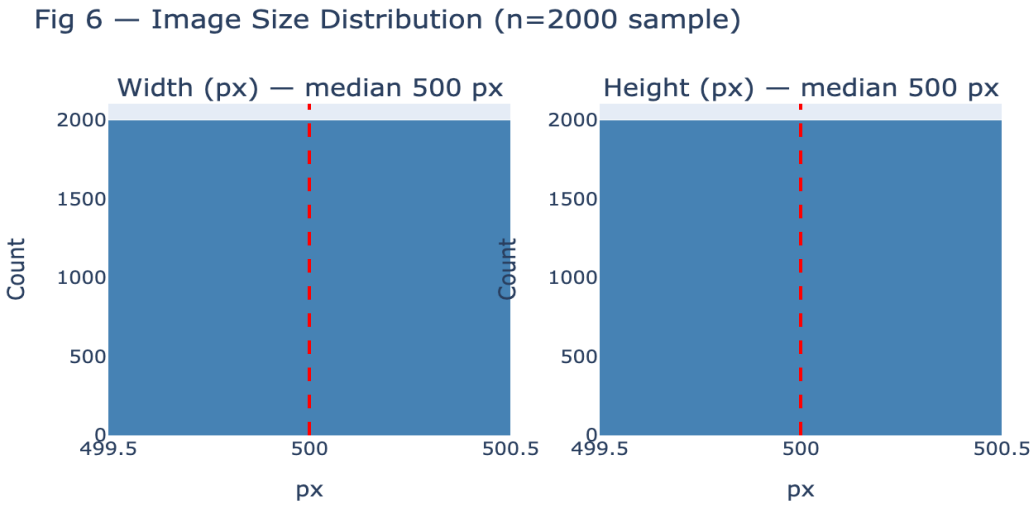


Fig 6 — Distribution of image widths and heights across the training set.

Business insight: All product images are uniformly 500×500 px (std = 0). Downsampling to 224×224 required for CNN input.

Recommendation:

- Resize all images to 224×224 px (ResNet, EfficientNet, ViT pretrained on ImageNet)
- Apply centre-crop rather than stretch resize to preserve aspect ratios
- Normalise with ImageNet mean/std: [0.485, 0.456, 0.406] / [0.229, 0.224, 0.225]

4. Preprocessing & Feature Engineering

4.1 Text Preprocessing Pipeline

- **HTML stripping** — regex removal of all HTML tags from description
- **Missing value handling** — NaN descriptions replaced with empty string
- **Unicode normalisation** — NFD decomposition + ASCII for accented French characters
- **Lowercasing + punctuation/digit removal**
- **Stopword removal** — French, English, German (NLTK)
- **Feature combination** — designation + cleaned description → text_clean

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Raw  : 'Olivia: Personalisiertes Notizbuch 150 Seiten Punktraster Ca Din A5...'
Clean: 'olivia personalisiertes notizbuch seiten punktraster ca din a rosen design'
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4.2 Engineered Features

Feature	Type	Description
text_clean	string	Cleaned designation + description concatenated
has_description	int (0/1)	1 if original description was non-empty
word_count	int	Word count of text_clean
char_count	int	Character count of text_clean

has_description rate: 64.9% of training rows have a non-empty description.

4.3 Image Preprocessing



Original 500×500 px (left) vs. 224×224 px resized (right).

- Target size for Rendu 2: **224×224 px** (ResNet / EfficientNet / ViT compatible)
- Normalisation: ImageNet mean [0.485, 0.456, 0.406] / std [0.229, 0.224, 0.225]
- Augmentation (training only, Rendu 2): random horizontal flip, colour jitter

4.4 Output Artefacts

File	Shape	Contents
data/train_cleaned.pkl	(84,916 × 9)	productid, imageid, designation, description, text_clean, has_description, word_count, char_count, prdtypecode
data/test_cleaned.pkl	(13,812 × 8)	Same columns minus prdtypecode

5. Conclusions

5.1 Key Findings

Finding	Detail
Class imbalance	Top class ~13× larger than smallest (χ^2 p ■ 0.05). Weighted loss required.
Missing descriptions	35.1% of rows — category-dependent (χ^2 p ■ 0.05). Image + designation are primary signals.
Text informativeness	Word count varies by category (Kruskal-Wallis p ■ 0.05). TF-IDF shows strong per-class vocabulary.
Missing rate vs size	Pearson r = –0.015 (p = 0.94): no link between missingness and category size.
Image uniformity	All images 500×500 px, no missing files. Resize to 224×224 for CNN.
Multilingual + HTML	Predominantly French; some German/English. HTML noise stripped.

5.2 Challenges for Rendu 2

- **Class imbalance** — class_weight='balanced' or weighted random sampling
- **Missing descriptions** — fallback to designation only; has_description flags this
- **Multimodal fusion** — text encoder + image encoder → late fusion
- **Image standardisation** — 224×224 px for CNN/ViT
- **French NLP** — CamemBERT or FlauBERT for best text representation

5.3 Modelling Roadmap (Rendu 2)

Stage	Approach
Baseline text	TF-IDF + Logistic Regression / Random Forest
Baseline image	ResNet50 pretrained on ImageNet, fine-tuned on product images
Advanced text	CamemBERT / FlauBERT fine-tuned for sequence classification
Advanced image	EfficientNet-B4 or ViT with data augmentation
Fusion	Late fusion: text + image softmax → MLP classifier
Evaluation	Weighted F1-score (macro) — competition metric